

From MLPs to Trainable Models

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Activations, objectives, and initialisation

Jonathon Hare and Antonia Marcu

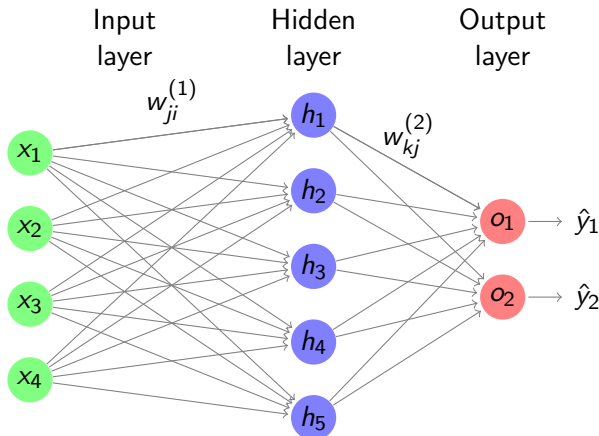
Vision, Learning and Control
University of Southampton

- Multilayer perceptrons as compositions of affine maps and nonlinearities
- Activations, logits, and loss functions
- The complete training pipeline
- Initialisation and the scale of signals and gradients
- Input scaling, weight initialisation, and internal normalisation

How Supervised Learning Typically Works

- Start by choosing a model-class: $\hat{y} = f(\mathbf{x}; \mathbf{W})$ where the model-class f is a way of using some numerical parameters, $\mathbf{W}$, to map each input vector $\mathbf{x}$ to a predicted output $\hat{y}$.
- *Learning* means adjusting the parameters to reduce the discrepancy between the true target output y on each training case and the output $\hat{y}$, predicted by the model.

Let's look at a Multilayer Perceptron (without biases)...



Without loss of generality, we can write the above as:

$$\hat{\mathbf{y}} = g(f(\mathbf{x}; \mathbf{W}^{(1)}); \mathbf{W}^{(2)}) = g(\mathbf{W}^{(2)} f(\mathbf{W}^{(1)} \mathbf{x}))$$

where f and g are activation functions.

Common Activation Functions

- Identity
- Sigmoid (aka Logistic)
- Hyperbolic Tangent ($\tanh$)
- Rectified Linear Unit (ReLU) (aka Threshold Linear)

Final layer activations

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- What form should the final layer function g take?
- It depends on the task (and on the chosen loss function)...
 - regression $\rightarrow$ typically linear
 - binary classification $\rightarrow$ typically Sigmoid
 - multilabel classification $\rightarrow$ typically Sigmoid
 - multiclass classification $\rightarrow$ typically Softmax

$$\text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}} \quad \forall i = 1, 2, \dots, K$$

- The outputs are positive and sum to 1.
- The denominator contains every logit, so every output depends on every input.
- Let $\mathbf{s} = \text{softmax}(\mathbf{z})$. The entries of its Jacobian are:

$$\frac{\partial s_i}{\partial z_j} = s_i(\delta_{ij} - s_j).$$

Here δ_{ij} is the Kronecker delta: it is 1 if $i = j$, and 0 otherwise.

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- Softmax replaces the hard one-hot vector with a continuous, differentiable distribution:

$$\begin{aligned} \mathbf{z} &= \begin{bmatrix} 1.1 & 4.0 & -0.1 & 2.3 \end{bmatrix} \\ \text{onehot}(\arg \max_k z_k) &= \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \\ \text{softmax}(\mathbf{z}) &= \begin{bmatrix} 0.044 & 0.797 & 0.013 & 0.146 \end{bmatrix} \end{aligned}$$

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- The largest component stays associated with the predicted class, but the output retains information about the alternatives.

Temperature controls the relaxation

$$\text{softmax}(\mathbf{z}/T)_i = \frac{e^{z_i/T}}{\sum_{j=1}^K e^{z_j/T}}, \quad T > 0.$$

- Lower temperatures sharpen the distribution:

$$T = 1.0 : \begin{bmatrix} 0.044 & 0.797 & 0.013 & 0.146 \end{bmatrix}$$

$$T = 0.6 : \begin{bmatrix} 0.008 & 0.937 & 0.001 & 0.055 \end{bmatrix}$$

$$T = 0.2 : \begin{bmatrix} 5.0 \times 10^{-7} & 0.9998 & 1.3 \times 10^{-9} & 0.0002 \end{bmatrix}$$

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- Higher temperatures flatten the distribution. As $T \rightarrow \infty$, it approaches the uniform distribution.

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- There are many different loss functions we might encounter (MSE, Cross-Entropy, KL-Divergence, huber, L1 (MAE), CTC, Triplet, ...) for different tasks.

The Loss Function (measure of discrepancy)

Recall from Foundations of Machine Learning:

- Mean Squared Error (MSE) loss for a single data point is given by

$$\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y}) = \sum_i (\hat{y}_i - y_i)^2 = (\hat{\mathbf{y}} - \mathbf{y})^\top (\hat{\mathbf{y}} - \mathbf{y})$$

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- $\ell_{MSE}(\hat{\mathbf{y}}, \mathbf{y})$ is the predominant choice for regression problems with linear activation in the last layer
- For a classification problem with Softmax or Sigmoidal activations MSE can cause slow learning
 - Gradients of ℓ_{MSE} are proportional to the difference in target and predicted value, multiplied by the gradient of the activation function
 - **The Cross-Entropy loss function is generally a better choice in this case**

Binary Cross-Entropy

For the binary classification case:

$$\ell_{BCE}(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

- The cross-entropy loss function is non-negative, $\ell_{BCE} > 0$
- $\ell_{BCE} \approx 0$ when the prediction and targets are equal (i.e. $y = 0$ and $\hat{y} = 0$ or when $y = 1$ and $\hat{y} = 1$)
- With Sigmoidal final layer, $\frac{\partial \ell_{BCE}}{\partial \mathbf{w}_i^{(2)}}$ is proportional to just the error in the output ($\hat{y} - y$) and therefore, the larger the error, the faster the network will learn!
- Note that the BCE is the negative log likelihood of the Bernoulli Distribution

Binary Cross-Entropy — Intuition

- The cross-entropy can be thought of as a **measure of surprise**.
- Given some input x_i , we can think of $\hat{y}_i$ as the estimated probability that x_i belongs to class 1, and $1 - \hat{y}_i$ is the estimated probability that it belongs to class 0.
- Note the extreme case of infinite cross-entropy, if your model believes that a class has 0 probability of occurrence, and yet the class appears in the data, the 'surprise' of your model will be infinitely great.

Binary Cross-Entropy for multiple labels

In the case of multi-label classification with a network with multiple sigmoidal outputs you just sum the BCE over the outputs:

$$\ell_{BCE} = - \sum_{k=1}^K [y_k \log(\hat{y}_k) + (1 - y_k) \log(1 - \hat{y}_k)]$$

where K is the number of classes of the classification problem, $\hat{y} \in \mathbb{R}^K$.

Numerical Stability: The Log-Sum-Exp trick

$$\ell_{BCE}(\hat{y}, y) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

- Consider what might happen early in training when the model might confidently predict a positive example as negative
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- To tackle this problem implementations usually combine the sigmoid computation and BCE into a single loss function that you would apply to a network with linear outputs (e.g. `BCEWithLogitsLoss`).
- Internally, a trick called 'log-sum-exp' is used to *shift* the centre of an exponential sum so that only numerical underflow can potentially happen, rather than overflow
- - Ultimately this means you'll always get a numerically reasonable result (and will avoid NaNs and Infs originating from this point).

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Log-Sum-Exp can be used for better numerical stability. PyTorch combines LogSoftmax with NLL in one loss and calls this “Categorical Cross-Entropy” (so you would use this with a **linear output layer**)

The complete training pipeline

- 1 Initialise the model parameters θ and the optimiser state.

One PyTorch training step

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optimizer.zero_grad() → loss = criterion(model(x), y) →  
loss.backward() → optimizer.step()
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- 4 Update the parameters and repeat with another minibatch.

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 - Biases may still be initialised to zero because the random weights have already broken the symmetry.
- With ReLU, zero pre-activations are especially problematic because PyTorch uses a derivative of zero at the origin.

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- If $c = 2$, the same network amplifies it by about 10^6 .
- Backpropagation has the same repeated-multiplication problem. We want activations and gradients to stay at a useful scale.

Variance propagation through a linear layer

For one output of a linear layer,

$$z_j = \sum_{i=1}^{n_{\text{in}}} w_{ji} x_i.$$

- Assume the inputs and weights are independent, zero-centred, and identically distributed.

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- Applying the same argument to the backward pass suggests $\sigma_w^2 \approx 1/n_{\text{out}}$.

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or

$$w \sim \mathcal{U}\left[-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right].$$

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- This reasoning fits linear activations and tanh near its approximately linear region.

Why Xavier under-scales ReLU networks

Let $q_\ell = \mathbb{E}[(h^{(\ell)})^2]$. If the pre-activations are symmetric around zero, then

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- Repeated ReLU layers can shrink the activation scale by roughly 2^{-L} . The backward pass loses a similar factor because only the active ReLUs transmit gradients.

He and Kaiming initialisation

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- Now the activation second moment is approximately preserved:

$$q_{\ell} \approx \frac{1}{2} n_{\text{in}} \frac{2}{n_{\text{in}}} q_{\ell-1} = q_{\ell-1}.$$

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- He initialisation compensates for the factor of $\frac{1}{2}$ introduced by ReLU:

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$$w \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}}}\right) \quad \text{or} \quad w \sim \mathcal{U}\left[-\sqrt{\frac{6}{n_{\text{in}}}}, \sqrt{\frac{6}{n_{\text{in}}}}\right].$$

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- The correct gain depends on the activation. Leaky ReLU, for example, also passes part of the negative signal.

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- Initialise explicitly when the activation assumption matters:

```
nn.init.kaiming_normal_(layer.weight,  
                        nonlinearity="relu")  
nn.init.zeros_(layer.bias)
```

Input scaling, initialisation, and internal normalisation

Mechanism	What it controls	When it acts
Input scaling	The scale and conditioning of the data entering the first layer	Before training. Estimate scaling statistics from the training data only.
Weight initialisation	The starting scale and symmetry of the parameters	Once, when the model is constructed. Match it to the activation.
Internal normalisation	The scale of learned activations inside the network	During each forward pass. BatchNorm and LayerNorm use different statistics.

- These mechanisms complement one another. Normalisation does not make careful data scaling or sensible initialisation irrelevant.
- We will return to internal normalisation in the practical training